

Here is a comprehensive list of the core variable definitions utilized throughout the Unified Master Architecture (UMA) and Recursive Symbiotic Logic System (RSLS) field theory, separated by their operational domains.

1. RSLS Field Theory & Differential Geometry

* ** $g_{\mu\nu}$ ** : The gravitational metric tensor defining the dynamical spacetime on the 4-dimensional pseudo-Riemannian manifold.[1]

* ** U^μ ** : The timelike vector flow (congruence) representing the kinematic state and causal propagation of information.[1]

* ** M ** : The thermodynamic memory field; a causal, dissipative scalar field acting as an Israel-Stewart state variable tracking local informational entropy.[1]

* ** M_{max} ** : The absolute saturation limit (thermodynamic capacity bound) of the memory field.[1]

* ** $V(M)$ ** : The strictly convex logarithmic barrier potential (the "Holographic Wall") preventing the memory field from exceeding its maximum limit.[1]

* ** α ** : The dynamic metric lapse function representing gravitational time dilation.[1]

* ** β^r / β^ϕ ** : The spatial shift vector representing radial information inflow or geometric frame-dragging.[1]

* ** ℓ ** : The emergent macroscopic relaxation length scale, establishing the physical thickness of the

diffuse-interface phase boundary (the entropy wall).[1, 1]

* ** τ_M ** : The thermodynamic relaxation time representing the causal lag between geometric forcing and informational response.[1]

* ** τ_J ** : The entropy transport relaxation time governing the Maxwell-Cattaneo telegrapher's equation limit.[1]

* ** μ ** : The intrinsic stiffness modulus, utilized as a parabolic spatial gradient penalty to enforce Bounded Variation (BV) closure across the phase boundary.[1]

* ** J_μ ** : The dynamically evolving entropy-gradient flux vector utilized to maintain strict Lorentz causality (subluminal transport).[1]

* ** θ ** : The geometric expansion scalar ($\nabla_\mu U^\mu$) tracking spatial volume changes.[1]

* ** κ ** : In the kinematic sector, this acts as the Lagrange multiplier enforcing the vector flow normalization ($U^\mu U_\mu = 1$).[1] In the Tensor Bridge, it acts as the proportionality constant derived via least squares ($\langle T | \rangle = \kappa \cdot G$) mapping the field to General Relativity.[1]

* ** $T_{\mu\nu}$ ** : The spatial Martin-Siggia-Rose (MSR) stress-energy tensor generated by the field.[1, 1]

* ** $G_{\mu\nu}^{(1)}$ ** : The linearized Einstein tensor.
[1]

* ** $h_{\mu\nu}$ ** : The spatial metric perturbation derived from the mass-energy component of the stress tensor via the Poisson equation.[1]

2. Acoustic Sphere & Hardware Geometry

* **R***: The internal physical radius of the acoustic sphere cavity.[1]

* **j_{nl}***: The exact zero (root) of the spherical Bessel function dictating the resonant spatial mode (e.g., $j_{01} = \pi$).[1]

* **k_{nl}***: The spatial wavenumber of the cavity mode, exacted as j_{nl}/R . [1]

* **f_{nl}***: The fundamental acoustic resonant frequency of the selected mode.[1]

* **L_{pend}***: The pendulum length, strictly derived to synchronize its period with the sphere's acoustic resonance.[1]

* **r_0***: The dynamic physical radius of the Venturi throat, constrained to sit at exact resonance with the Bessel zeros.[1]

* **r_{in}***: The fixed inner cutoff radius of the Venturi, derived from the preceding Bessel zero ($j_{(n-1)l}$).[1]

* **G***: The Bernoulli gain, establishing the fluidic injection coupling strength strictly from the ratio of r_{in} to r_0 . [1, 1]

* **d***: The spacing of the blind gate acting as a diffraction grating to produce a first-order optical maximum.[1]

* **f_{fan}***: The mechanical fan modulation frequency.[1]

* **T_{star}***: The exact stellar temperature (in Kelvin)

that the closed system geometry is perfectly resonant with.[1]

3. Semantic Engine & Computational Variables

* ** ψ ** : The primary continuous complex field state matrix evolving under GENERIC thermodynamic dynamics.[1]

* ** $\hat{\psi}$ ** : The exact MSR response field conjugate required for computing dissipative drift, calculated mathematically as $-dH/d\psi^*$.[1, 1]

* ** z ** : The projected lower-dimensional state vector mapping the continuous field into the bounded execution Fourier space.[1, 1]

* ** x ** : The discrete binary weight scalar calculated from tokenizing semantic text inputs relative to an anchor string.[1]

* ** z_c ** : The Engine3 complex state seed mapping the discrete binary weight into a continuous spatial amplitude and phase.[1, 1]

* ** $H[\psi]$ ** : The actual GENERIC Hamiltonian energy of the field utilized for measuring semantic friction (tracking dH/dt).[1]

* ** E ** : The historical DC mode energy proxy, calculated as $|z|^2/2$.[1]

* ** E^* ** : The calibrated thermodynamic target energy the Venturi auto-regulates the field toward.[1]

* ** dt ** : The discrete computational time step applied to the forward evolution operators.[1]

* ** tol ** : The generalized mathematical tolerance

bound utilized across solvers. Applied as `metric_tol` to break the self-consistent metric iteration loop, or `closure_state_tol` to guarantee spatial deviation boundaries are satisfied prior to declaring semantic closure.[1, 1, 1]

* **Q**: A positive semi-definite matrix defining the ellipsoidal volume metric in Constraint Gamma, specifically encoding Fisher information geometry to bound data density.[1]

* **\Sigma**: The field's covariance matrix used to calculate the macroscopic scaling coefficient (information concentration) upon successful state-space closure.[1]

* **W**: A calibrated linear readout matrix utilized by the Projection Inarticulator to map the high-dimensional closed mathematical state into specific, low-dimensional production metrics.[1]